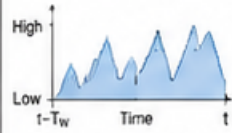


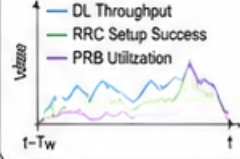
1. Input Observation Window

 Serving and Neighboring Cells
+ Operational Context


UE / Load Activity



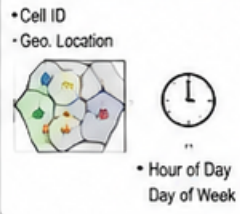
KPI Monitoring (Examples)



OSS Alarm Events

Time	Alarm ID	Cell	Severity
t-5	10021	1	Major
t-17	30045	2	Minor
t-21	20011	1	Major
...	...	...	...
t-2	40033	3	Minor

Cell / Time Information



2. Feature Representation Layer

KPI Patch Tokens

 Segmented time-series
representation


Alarm Event Tokens

 Sparse event sequence
representation


Cell / Time Positional Features

 Spatial-temporal context
representation


Event Intensity / Influence Features

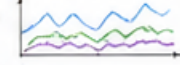
Weighted relationship graph



3. CET-STGL Representation Learning

Causal-inspired influence modeling

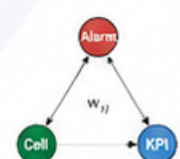
1) KPI Temporal Encoding

 Capture temporal dynamics
of multi-KPI sequences


2) Event Encoding

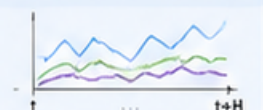
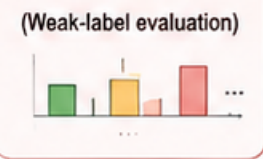
 Encode sparse alarm events
and severity patterns

 3) Alarm-RAN Cell-KPI
Influence Graph

 Learn influence weights
among alarms, cells,
and KPIs

 4) Spatio-Temporal
Cross-Cell Graph

 Model cross-cell
dependencies with
temporal context


4. Reliability Task Heads

 KPI Forecasting
(Real future KPI values)

 Weak Degradation
Classification
(Weak-label evaluation)

 Weak Root-Cause
Candidate Ranking
(Proxy target)


Reported Experimental Implementation

Same Feature Blocks

KPI Patch Tokens



Alarm Event Tokens



Cell / Time Positional Features



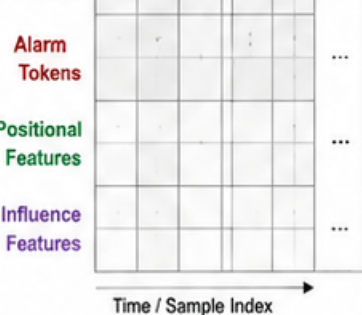
Event Intensity / Influence Features



Feature Matrix Construction

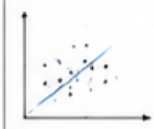
 Concatenate and align features
over the observation window

KPI Tokens



Sklearn Proxy Models

Ridge Regression



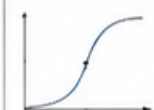
- Linear model
- L2 regularization

Random Forest



- Non-linear model
- Ensemble of decision trees

Logistic Regression



- Classification model
- Probabilistic output

Evaluation Metrics

- KPI Forecasting
 - MAE / RMSE / MAPE
- Weak Degradation Classification
 - Accuracy / F1-score / AUC
- Weak Root-Cause Ranking
 - Hit@K / MRR / NDCG

 Full neural spatio-temporal
graph backend remains
conceptual.